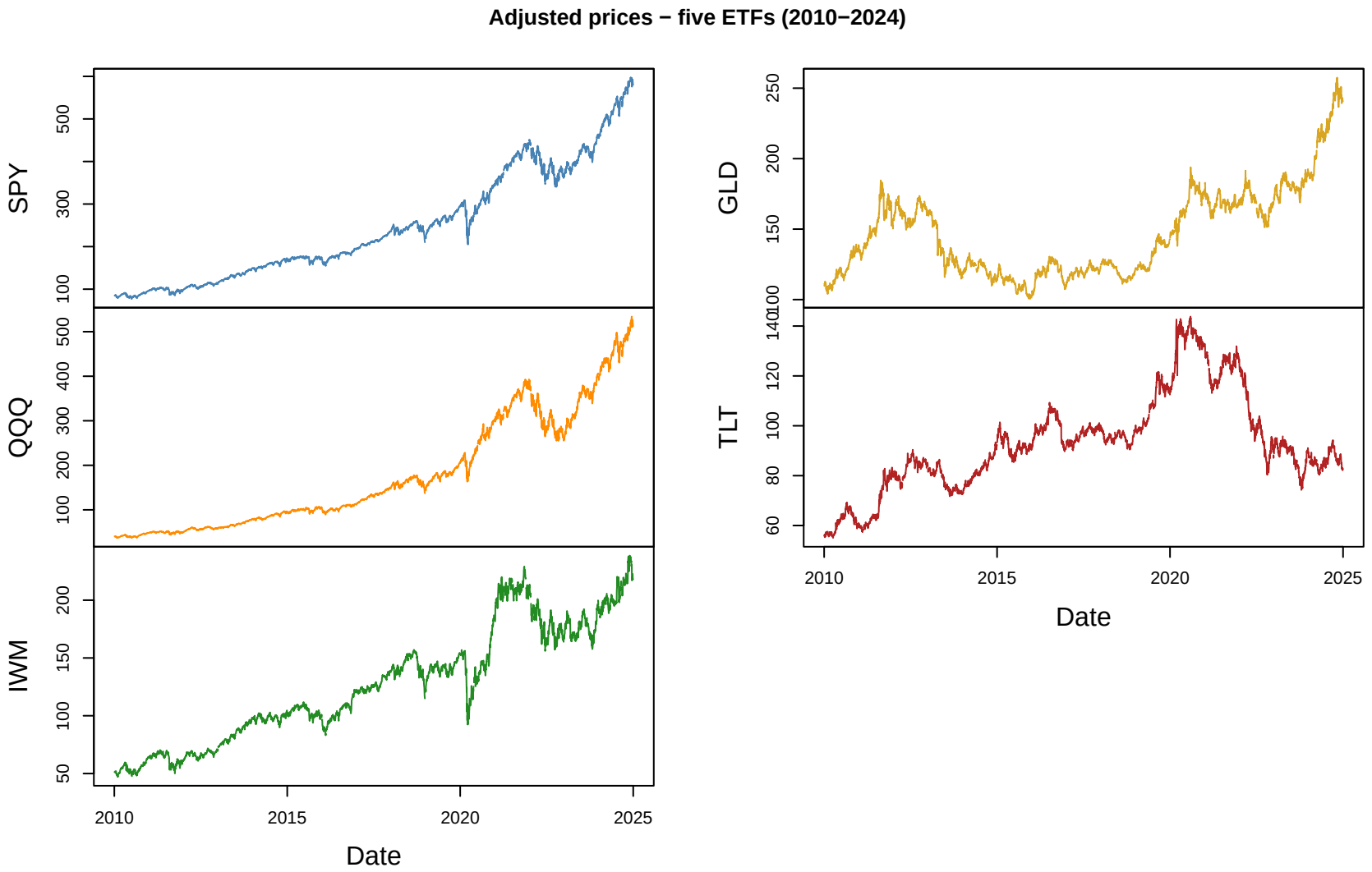
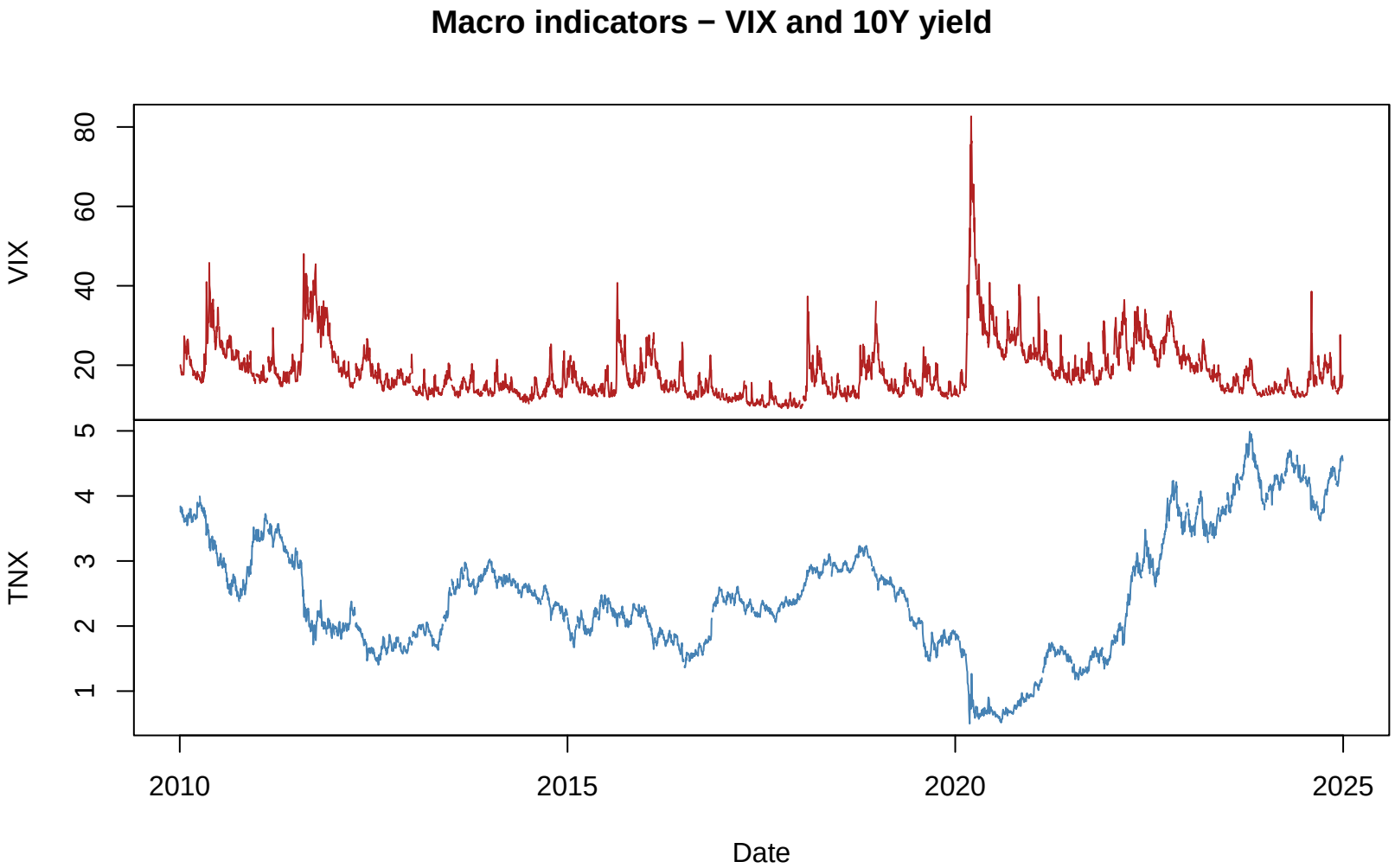
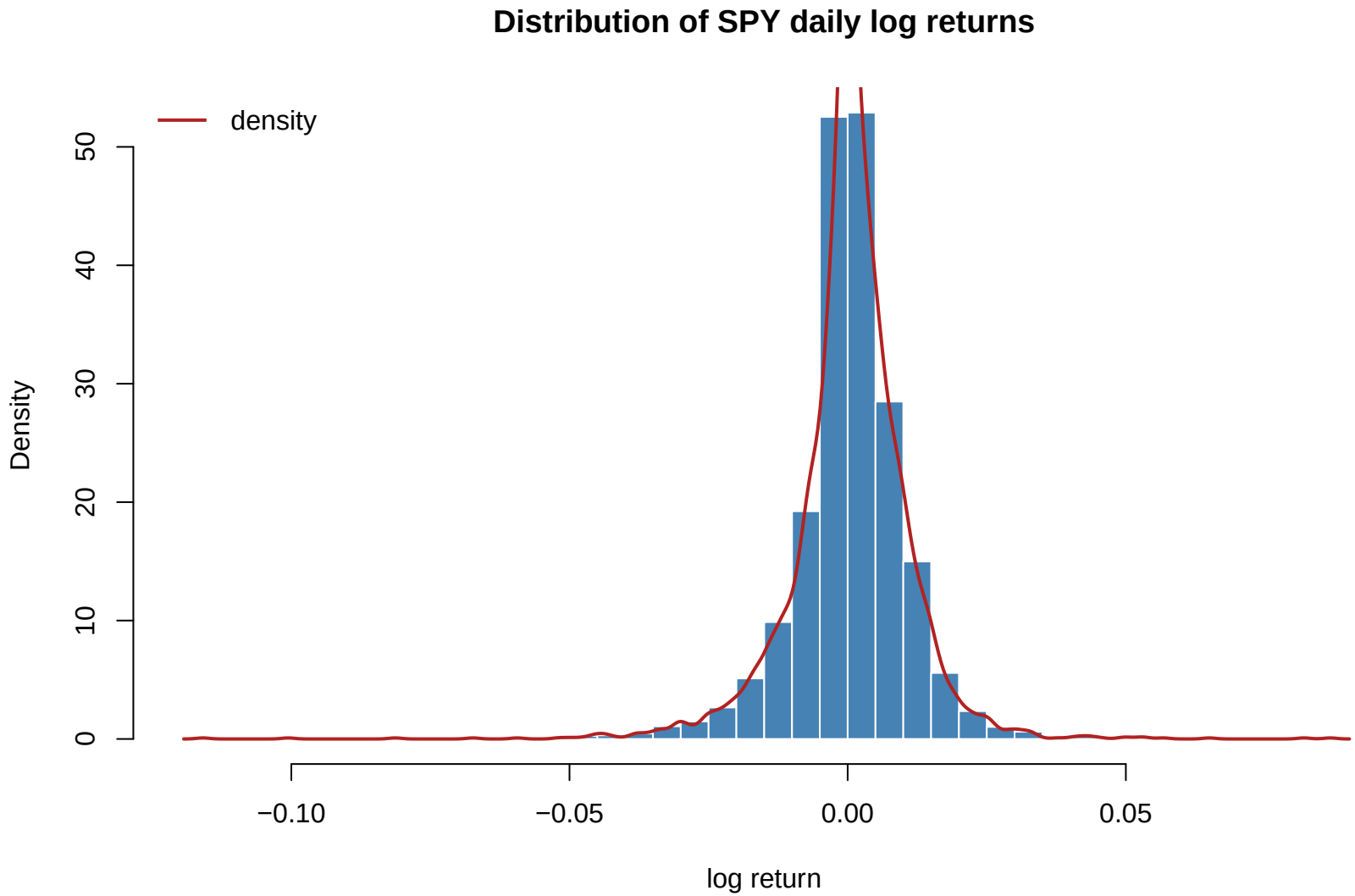
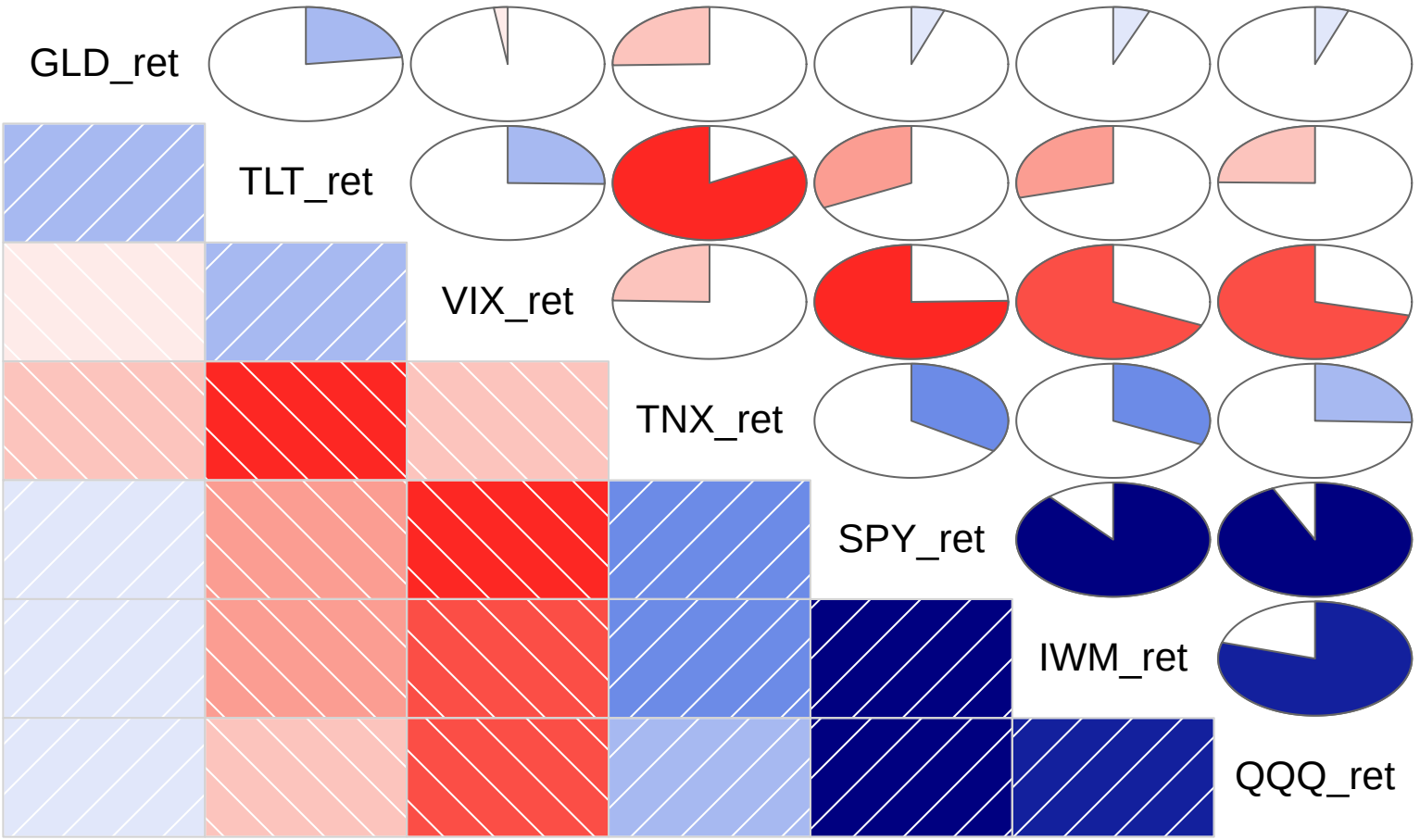

Adjusted prices − five ETFs (2010−2024)
SPY
QQQ
IWM
GLD
TLT
Date
Date
2010
2015
2020
2025

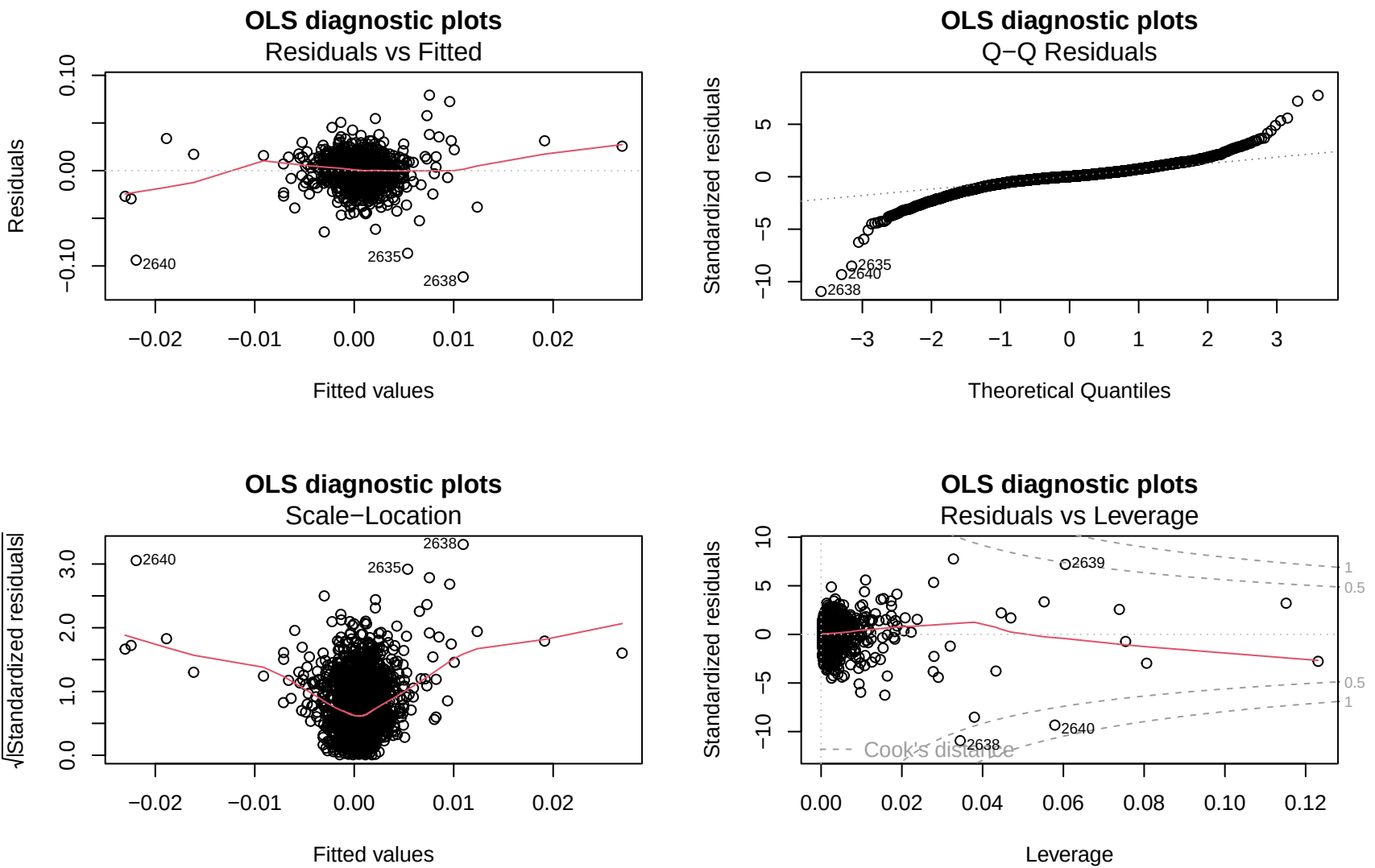

Macro indicators − VIX and 10Y yield
VIX
80
60
40
20
TNX
5
4
3
2
1
2010
2015
2020
2025
Date

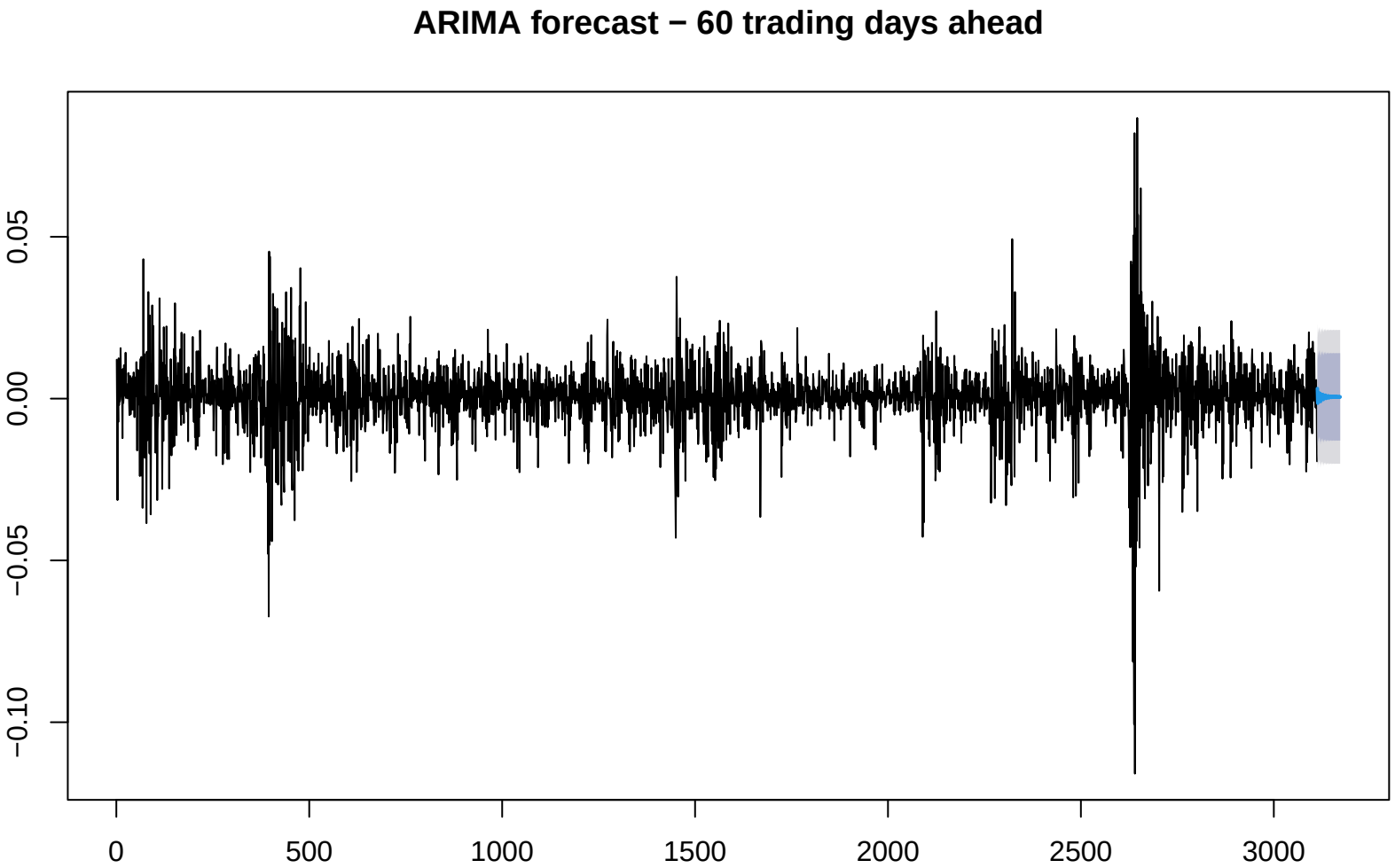

Distribution of SPY daily log returns
density
Density
0
10
20
30
40
50
−0.10
−0.05
0.00
0.05
log return

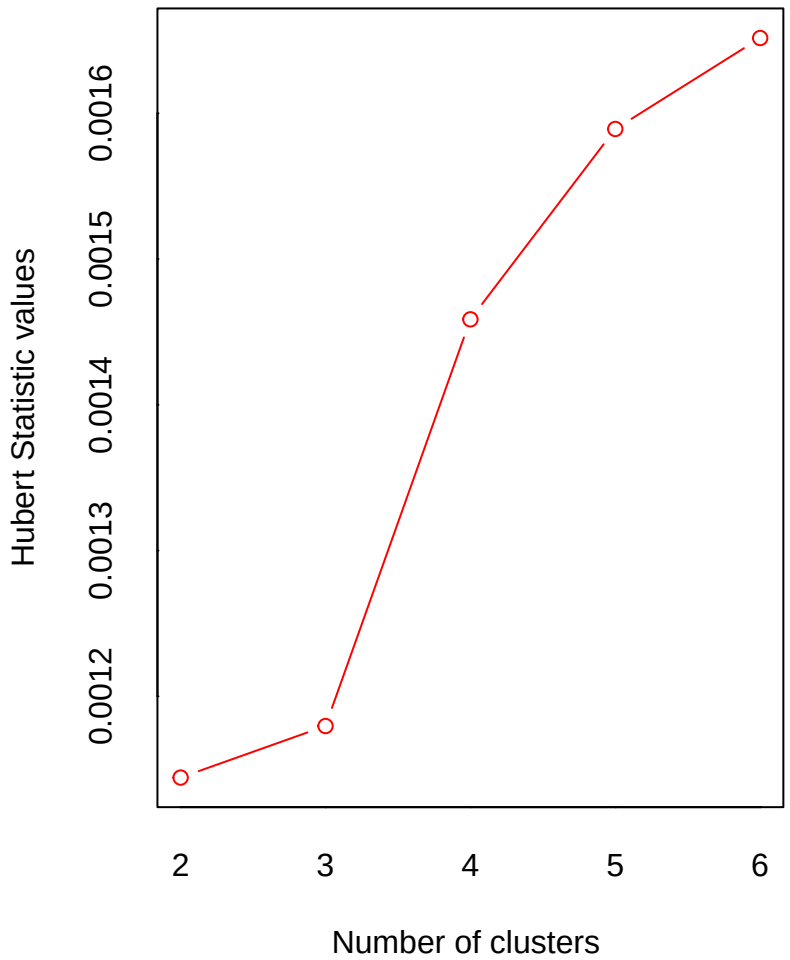
Cross−asset return correlations

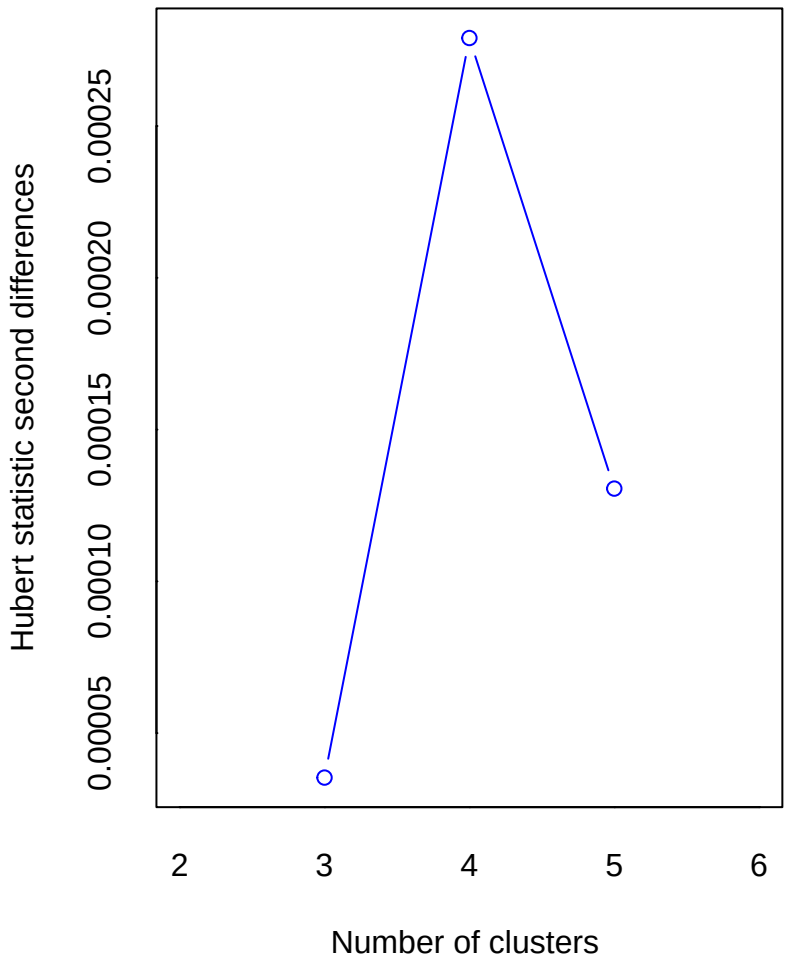


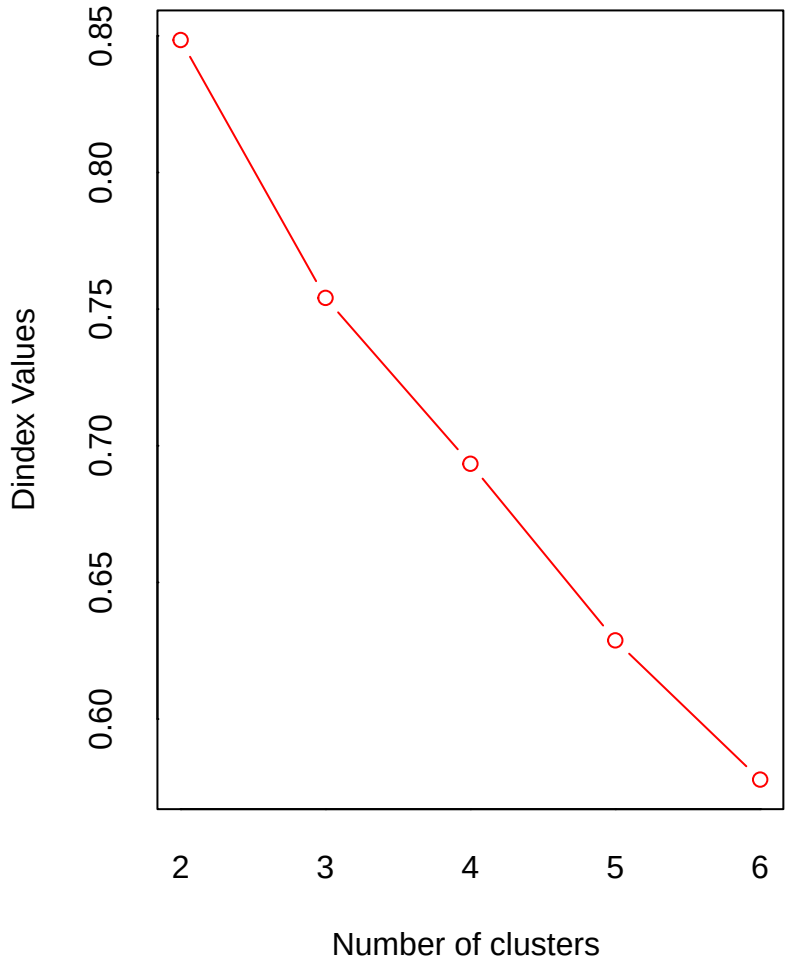
SPY cumulative log return
2010−01−05 / 2024−12−30
4
3
2
1
0
2010−01−05
2012−04−02
2014−07−01
2016−10−03
2019−01−01
2021−04−01
2023−07−03

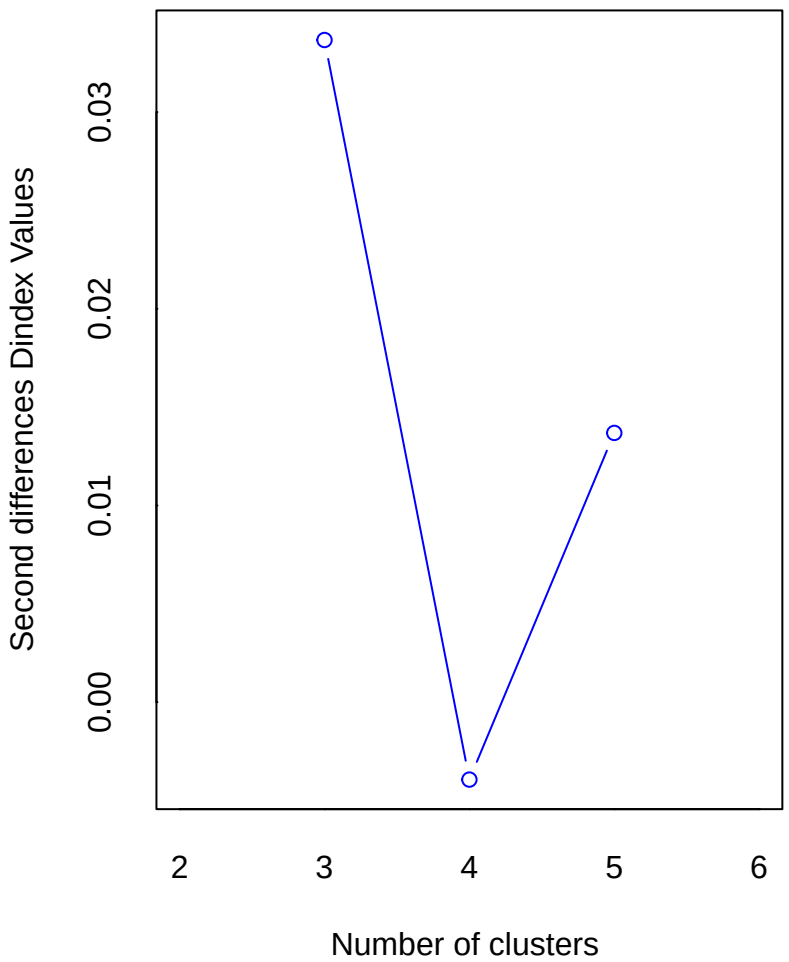

OLS diagnostic plots
Residuals vs Fitted
Residuals
Fitted values
2640
2635
2638
OLS diagnostic plots
Q−Q Residuals
Standardized residuals
Theoretical Quantiles
2635
2640
2638
OLS diagnostic plots
Scale−Location
√|Standardized residuals|
Fitted values
2640
2638
2635
OLS diagnostic plots
Residuals vs Leverage
Standardized residuals
Leverage
2639
2640
2638
Cook's distance

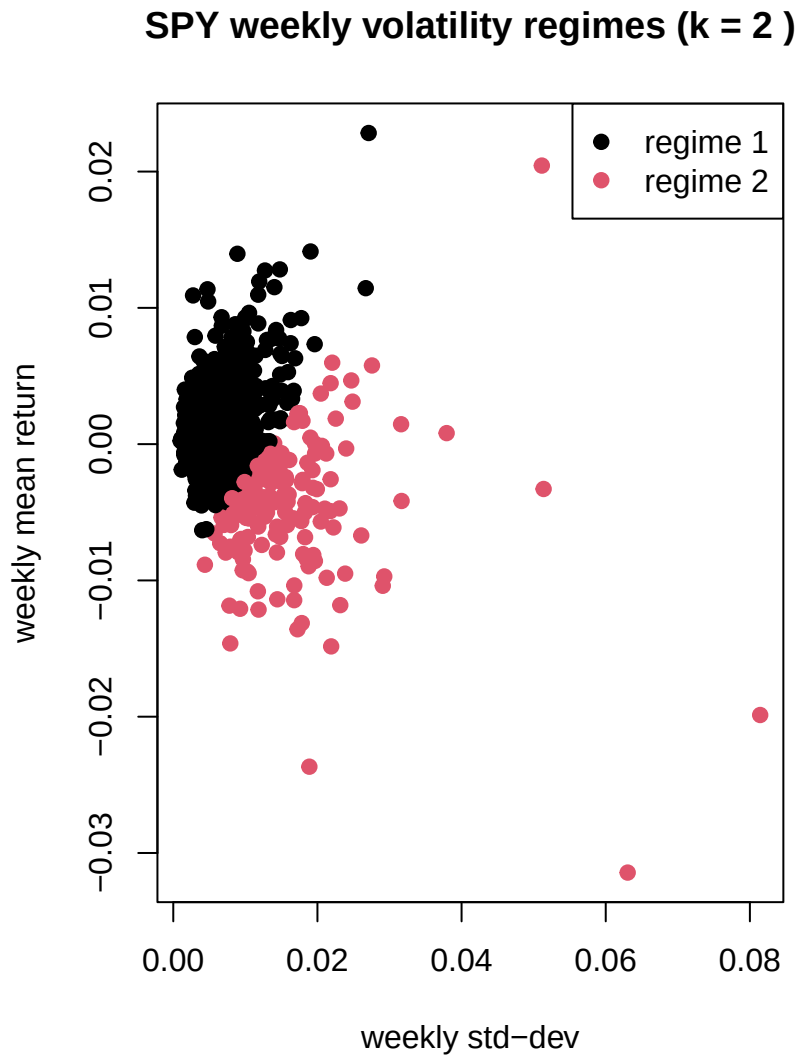

ARIMA forecast − 60 trading days ahead
0.05
0.00
−0.05
−0.10
0
500
1000
1500
2000
2500
3000

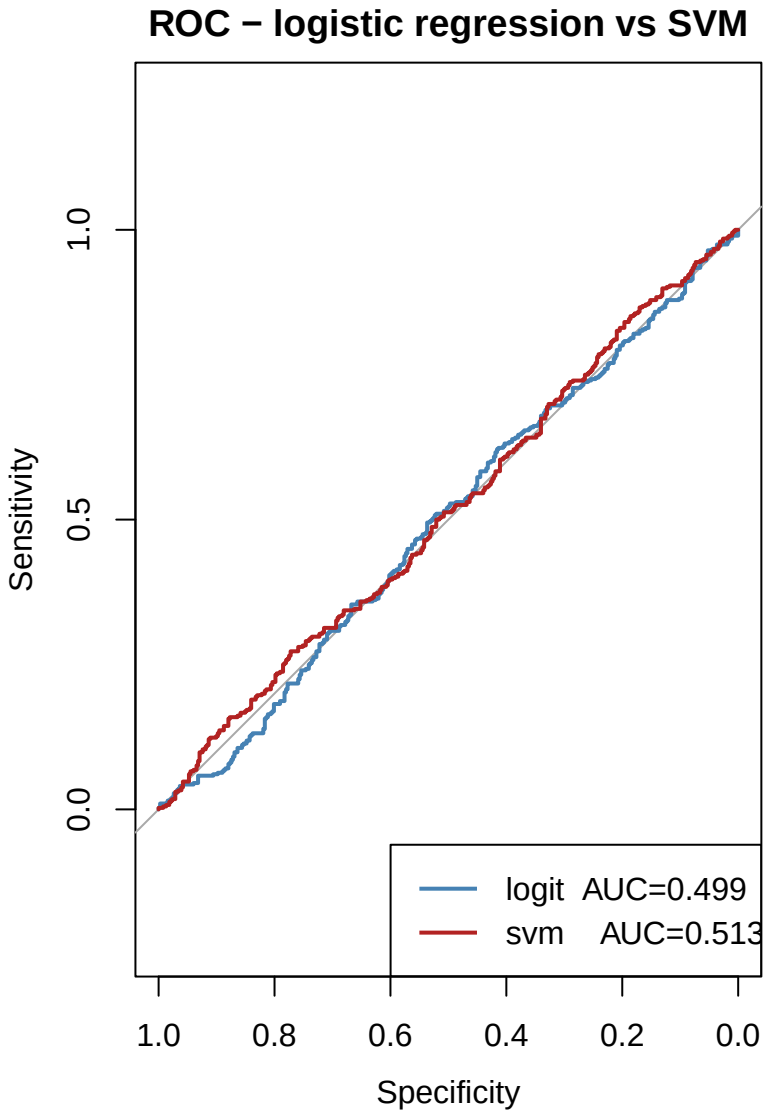

Hubert Statistic values
0.0012
0.0013
0.0014
0.0015
0.0016
2
3
4
5
6
Number of clusters


Hubert statistic second differences
0.00005
0.00010
0.00015
0.00020
0.00025
2
3
4
5
6
Number of clusters


Dindex Values
0.60
0.65
0.70
0.75
0.80
0.85
2
3
4
5
6
Number of clusters


Second differences Dindex Values
0.00
0.01
0.02
0.03
2
3
4
5
6
Number of clusters


SPY weekly volatility regimes (k = 2)
regime 1
regime 2
weekly mean return
0.02
0.01
0.00
−0.01
−0.02
−0.03
0.00
0.02
0.04
0.06
0.08
weekly std−dev


ROC − logistic regression vs SVM
Sensitivity
1.0
0.5
0.0
logit AUC=0.499
svm AUC=0.513
1.0
0.8
0.6
0.4
0.2
0.0
Specificity